

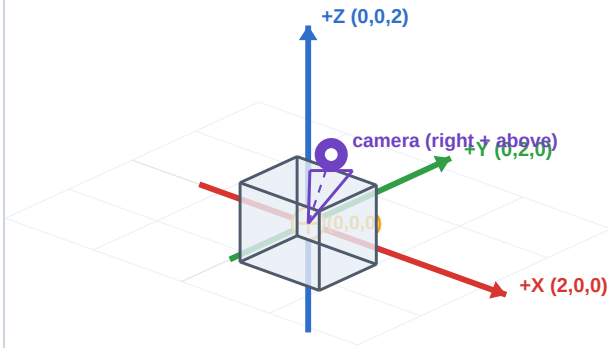
Blender -> Bevy Coordinate Plane

One-page sketch for visualizing origin, axes, cube placement, camera placement, and the up-axis conversion.

— X — Y — Z

1. Blender default world

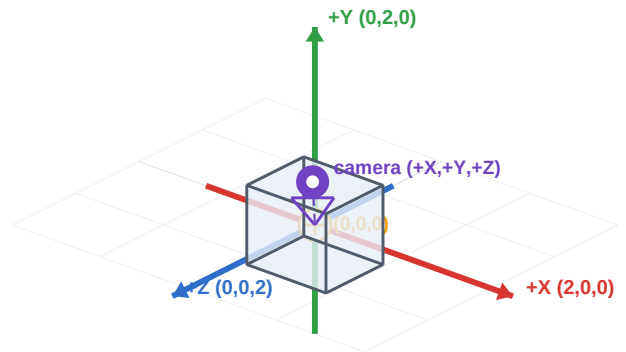
Right-handed, Z-up. Axis colors match the Blender viewport.



grid: X-Y floor, Z is up

2. Bevy default world

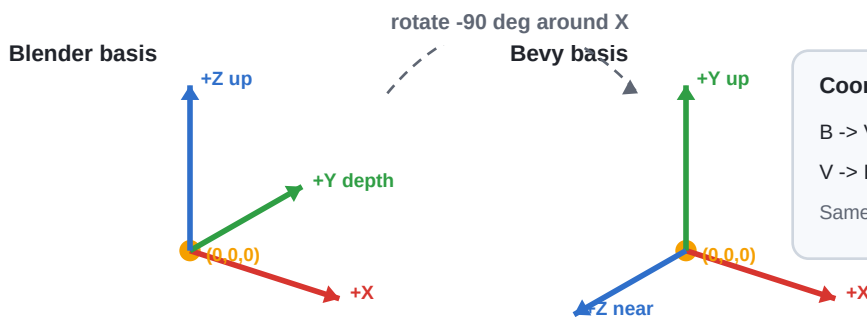
Right-handed, Y-up. Colors here reuse Blender's X/Y/Z colors.



grid: X-Z floor, Y is up

3. Mental rotation between them

Keep red X pinned. Move Blender's blue Z-up axis into Bevy's green Y-up axis.



Coordinate conversion

B -> V: $(x, y, z) \rightarrow (x, z, -y)$

V -> B: $(x, y, z) \rightarrow (x, -z, y)$

Same red X. Blender Z-up becomes Bevy Y-up.

Quick diagnostic

Sideways model: rotate -90 deg around X.

Backward model: flip forward, not the up axis.

How to read the sheet

Blender: X is left/right, Y is the front/back depth axis, Z is vertical. The default grid you model on is X-Y, with the cube centered at (0,0,0).

Bevy: X is right, Y is up, and +Z is toward the viewer. A 3D camera commonly sits at positive Z and looks back toward the origin along -Z.

The useful memory hook is simple: keep red X fixed, turn Blender Z-up into Bevy Y-up, then make Bevy +Z point opposite Blender +Y.

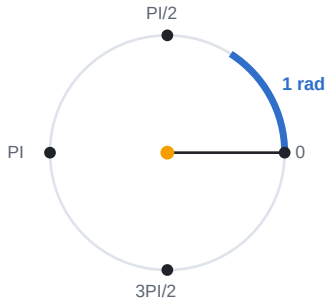
Rotation: Angles, Time & rotate_y

How angles, frame time, and a Y-axis spin actually work under transform.rotate_y.

— X — Y — Z

1. Angles: degrees, radians, TAU

Radians are the natural unit; one full turn = TAU = 2π .



Key turns

TAU = 2π = 6.28

PI = 3.14 (half)

PI/2 = 1.57 (qtr)

Think in turns: $0.1 * \text{TAU}$ reads as a tenth of a turn, far clearer than 0.628 radians. Use TAU for spins, PI for half-flips; convert human degrees at the edge with `to_radians()`.

2. Hz, frames & delta time

Scale motion by delta so the speed is frame-rate independent.



rotate_y(0.1 * TAU * dt)

0.1 = turns per second (0.1 Hz)

TAU = turns -> radians

dt = this frame's slice of a second

angular speed $w = \text{TAU} * f$

3. Under the hood: rebuilding rotate_y

looking straight down +Y (top view)

Rotating about Y leaves Y fixed and spins X and Z in the plane -- the 2D rotation, applied to X and Z.

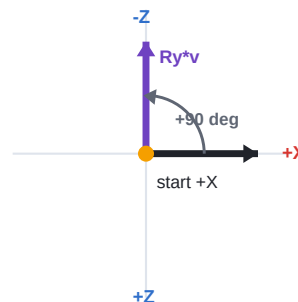
$$x' = x \cos t + z \sin t$$

$$y' = y \quad (\text{axis stays})$$

$$z' = -x \sin t + z \cos t$$

glam ships this as `Mat3::from_rotation_y(t)`, or the quaternion `Quat::from_rotation_y(t) = (0, \sin(t/2), 0, \cos(t/2))`.

`rotate_y` composes it about the world Y; `rotate_local_y` spins about the object's own Y. (t = the angle, in radians.)



Ry(t)

$$\begin{bmatrix} \cos t & 0 & \sin t \\ 0 & 1 & 0 \\ -\sin t & 0 & \cos t \end{bmatrix}$$

quarter turn: +X -> -Z

Reading the turntable line

```
transform.rotate_y(0.1 * TAU * time.delta_secs());
```

0.1 * TAU is a tenth of a turn, so one full lap takes ten seconds.

* `delta_secs()` keeps the speed identical at 60 or 120 fps. Drop it and a 120 Hz screen spins about 120x too fast.

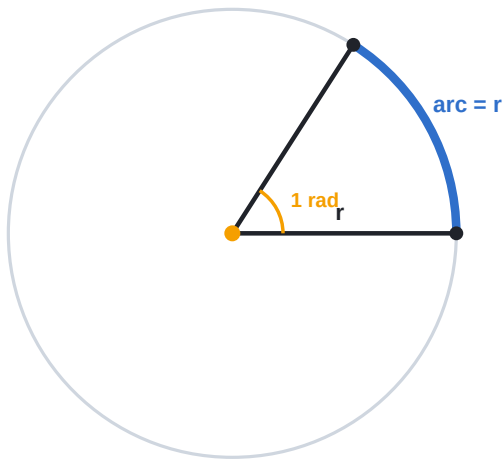
`rotate_y` spins it upright, like a turntable. `rotate_x` or `rotate_z` make it tumble instead.

What a Radian Really Is

A radian is a walk around the rim, measured in radius-lengths -- the natural way to size a turn.

1. One radian = one radius, walked

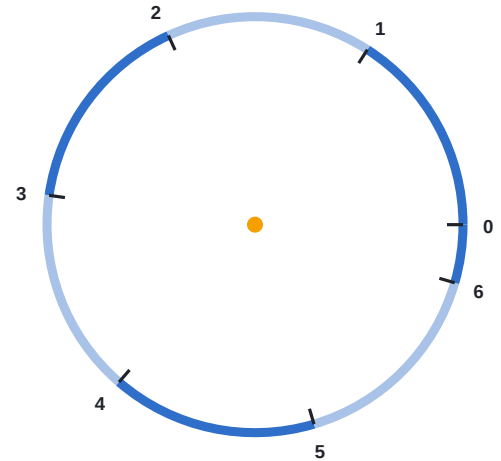
Bend the radius r onto the rim. The angle it covers is one radian.



The distance from the centre to the edge is the radius r . Walk that exact length along the curved edge, and the angle you have swept is ONE radian (about 57.3 degrees). No 360, no magic number -- just the radius laid onto its own rim.

2. Why a full turn is TAU

Keep walking radius-lengths around the rim.



Each coloured chunk of rim is one radius long = one radian. It takes $2\pi = 6.28$ of them to get all the way around. That is why one full turn = TAU = 2π radians, and a half turn = π .

Radians vs degrees -- and why the difference matters

Degrees chop a turn into 360 equal slices -- a number the Babylonians chose (near the days in a year). It has nothing to do with the circle itself.

Radians measure the actual walk along the rim, counted in radius-lengths. One radius walked = one radian ~ 57.3 deg. The whole rim is $2\pi \sim 6.28$ radii, so 360 deg = TAU.

The payoff: arc length = $r \times \text{angle}$, and $\sin / \cos$ come out clean -- but only when the angle is in radians. That is why Bevy, glam, and WGSL all expect radians, never degrees.

$$180 \text{ deg} = \pi$$

$$360 \text{ deg} = \text{TAU}$$

$$1 \text{ rad} \sim 57.3 \text{ deg}$$

$$\text{deg} \rightarrow \text{rad}: \times \pi/180$$